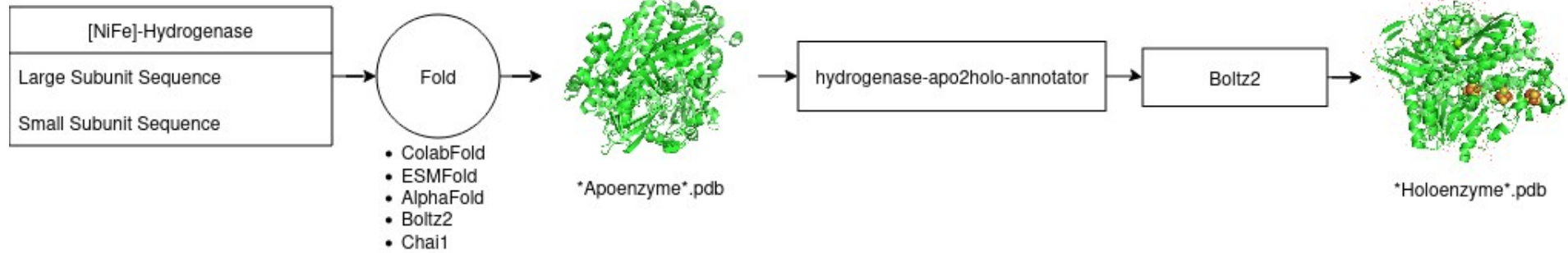


Hydrogenase-holoenzyme-builder pipeline



Apo2holo-annotator Part 1

1. **Search** empty *apo.pdb* **against** structure **database**
(foldseek easy-search)
2. **Filter** results (bitscore etc.)
3. **Structurally align** hits to query (*apo.pdb*)
4. **Identify** possible binding **pockets**

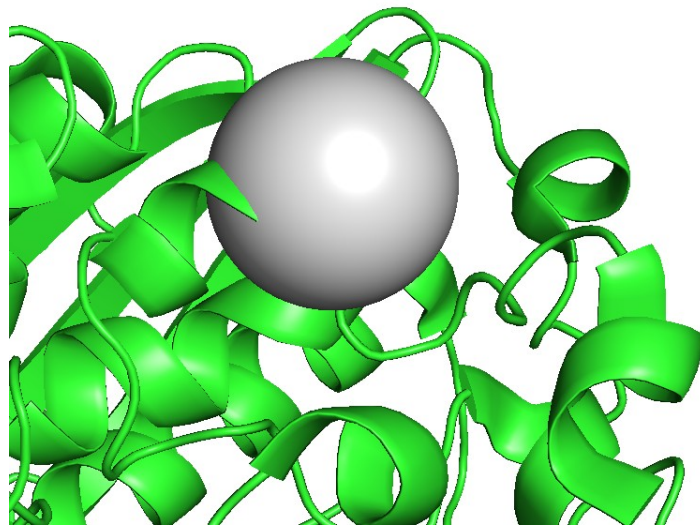
Apo2holo-annotator Part 2

For each target cofactor position in pocket

1. **Create** Aminoacid-**Fingerprint**
2. **Search** against FingerprintDB (*absolut* or *logreg*)
3. **Summarize** results

Aminoacid Fingerprint

- Count all aminoacids in radius 5 Å around point x,y,z



Apo2holo-annotator Part 3

For each pocket

1. **Select** Cofactor with highest confidence
2. **Identify cysteines** around binding pocket
3. **Write** output (*json* or *yaml* for boltz2)